User Sync Project Documentation

PeopleStrong to Microsoft Entra ID

This document describes the project as a business and operational system rather than a code walkthrough. It explains the purpose of the solution, the business value it provides, the integration flow between PeopleStrong and Microsoft Entra ID, deployment expectations, security considerations, and the operational practices required to run the platform successfully in a real enterprise environment.

1. Project Overview

The User Sync project is an enterprise identity synchronization solution designed to keep workforce data consistent between PeopleStrong HRIS and Microsoft Entra ID. It automates the retrieval of employee information, normalizes the structure to a stable internal representation, and then synchronizes the resulting user data to Microsoft Graph. This reduces manual data entry, keeps identities aligned with HR status changes, and gives administrators a structured way to monitor the health of the sync process.

2. Business Purpose

The system exists to solve a common organizational problem: when employee records change in the HR platform, the corresponding Microsoft Entra user account must also be updated. Without this automation, HR teams are forced to maintain separate records across systems, leading to stale profiles, inconsistent naming, mismatched department values, and delayed access changes for employees. This project acts as the bridge between PeopleStrong and Microsoft Entra ID, ensuring employee lifecycle events are reflected consistently.

3. Solution Components

- PeopleStrong HRIS connector: authenticates to PeopleStrong and fetches employee master records.
- Normalization layer: standardizes raw HR data into a structured employee format used by the application.
- Microsoft Graph integration: creates, updates, disables, and reviews Microsoft Entra users.
- Database tracking layer: stores employee synchronization status, logs, and audit context in SQLite.
- Admin dashboard: provides operational visibility, run summaries, and retry actions for admins.

4. High-Level Architecture

At a high level, the system is positioned between the PeopleStrong HR source and the Microsoft Entra target. PeopleStrong exposes employee master data through OAuth and a configured API endpoint. The Django app retrieves that data, filters it, normalizes it, and either stores it locally or forwards it to Microsoft Graph. Every step is recorded for both compliance and operational troubleshooting. This creates a traceable process from HR change to identity change.

5. Sync Flow

The full synchronization workflow proceeds in a consistent cycle. First, a sync job is started through a Django management command or a scheduled task. Second, the app requests an access token from PeopleStrong and calls the outbound HRIS API. Third, the employee payload is normalized into a form the application can use reliably. Fourth, the app checks whether the employee is eligible for Microsoft sync based on employment status. Fifth, Microsoft Graph performs the required create, update, or disable action. Finally, the outcome is persisted in the database together with execution details, making the operation audit-ready.

6. Employment Status Rules

One important governance rule in the project is that only employees whose status is Active or Inactive are sent to Microsoft for account update. Other values such as Probation, Notice Period, Resigned, Suspended, or an empty value are treated as ineligible for direct synchronization. Instead of risking the wrong lifecycle action, the application stores the record locally and logs a reason for skipping the sync. This is a critical safeguard for enterprise identity administration because it prevents accidental creation or reactivation of users in inappropriate states.

7. Data Model and Audit Trail

The project uses a small but important data model to support operational control. EmployeeSyncRecord stores the individual employee record and status details. SyncRun groups the run as a whole, and SyncLog provides granular event-level detail. This arrangement gives the platform a durable audit trail showing what happened, when it happened, and which employee it affected. It is especially valuable when administrators need to investigate failed transforms, retries, or Graph-side errors.

8. Operational Dashboard

The dashboard is designed for operational administrators rather than just developers. It presents summary metrics such as created, updated, deactivated, failed, and skipped records. It also allows staff to search, filter, and inspect recent run logs. From the UI, an employee can be retried, deactivated, or manually reviewed. This transforms the project from a background processor into an operational business tool that can be used in day-to-day identity maintenance.

9. Security and Permissions

This solution interacts with sensitive employee information and user lifecycle operations. It therefore requires secure handling of credentials, proper environment isolation, and careful permission setup. PeopleStrong requires valid OAuth credentials and API access; Microsoft Entra ID requires app registration credentials and the correct application permissions, typically with Graph rights that allow user read/write operations. Because the system can create, update, and disable accounts, administrative consent and least-privilege access are essential to maintain a safe production posture.

10. Configuration Requirements

- PeopleStrong configuration: token URL, API URL, API key, client ID, client secret, and integration name.
- Microsoft Entra configuration: tenant ID, client ID, client secret, and Graph permissions.
- Operational configuration: sync enable/disable flags, log file location, log level, and default company name.
- Environment isolation: a .env file or secure environment variables for each deployment environment.

11. Integration Risks and Controls

Any HR-to-identity integration carries operational risk. Mismatched field mappings, skipped statuses, expired credentials, permission changes, or network timeout conditions can all disrupt synchronization. The project mitigates these risks by maintaining run logs, storing raw and transformed payloads, recording failed record states, and exposing an admin retry workflow. These controls reduce the chance that a silent mismatch goes unnoticed.

12. Deployment Modes

The system works in both local and production-like deployment patterns. In a standard enterprise setup, it may be run from a Django environment on a server and scheduled through cron, Task Scheduler, or a hosted worker process. The application can also be run on a local workstation for testing and configuration validation. In all cases, the project should be deployed with a clear operational process for backups, monitoring, and credential rotation.

13. Operational Best Practices

- Validate environment variables before deployment and after credential rotation.
- Review sync logs after every scheduled run to detect errors early.
- Use retry-only runs for failed records instead of reprocessing everything.
- Keep the employment-status filter aligned with HR policy to avoid unintended account changes.
- Verify Graph permissions and consent before using the project in production.

14. Final Outcome

This project enables a disciplined employee identity synchronization process between PeopleStrong and Microsoft Entra ID. It supports business continuity, reduces manual administrative effort, provides clear auditability, and helps organizations maintain a consistent and trustworthy employee identity lifecycle across HR and Microsoft 365 platforms.

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